

Taylor Bosier

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Summary

Lead Data Scientist, 7+ years building production ML, Bayesian inference and optimization models that operational teams run daily — across warehousing, freight and medical devices. Stochastic and deterministic optimization, hierarchical Bayesian inference, causal measurement, and retrieval over large document estates.

Professional Experience

Lead Data Scientist Stryker (Remote)

Aug 2024 – Present

- Built a causal inference pipeline using Databricks SQL and Python to evaluate rebate ROI with propensity score matching and inverse propensity weighting across customer-level rebate engagement and eligible-product revenue.
- Developed a long-horizon forecasting pipeline (several XGBoost models) used for strategic planning of 18-36 month revenue forecasting, with cold start handling.
- Built a Databricks Vector Search workflow that parallelized ingestion of 35,000+ contracts from Azure Data Lake Storage, converted PDFs into chunked text, used AI-rewritten semantic queries to retrieve rebate structures, tied complex rebate terms to sales data, and rendered rebate attainment progress in a Next.js graph.
- Developed a pricing benchmark engine using XGBoost trained on historical sales, producing median and 75th-percentile price estimation by item/customer/region/contract type to standardize “market rate” guidance and reflect top-performer pricing behavior.
- Mentored senior data scientist and full-stack web developers and collaborated cross-functionally with finance, operations, and pricing teams to operationalize models.

Data Scientist II Knight-Swift Transportation, Phoenix, AZ

Sept 2021 – Aug 2024

- Promoted to Data Scientist II in Sept 2023.
- Re-architected and deployed a scheduling optimization model (Gurobi) for driver planning, raising planned drivers at one terminal from 20% to 85% and improving on-time delivery and network design.
- Designed an early-warning detection model surfacing 40+ drivers a week for safety remediation.
- Conducted a study using a Bayesian Poisson-Gamma model to quantify the safety impact of lane-keeping sensors, providing critical evidence for a multi-million dollar fleet hardware decision.
- Rebuilt a driver retention model, taking F1 on the churn class from 0.20 to 0.40; the model fed the retention program that ran alongside a 12% fall in driver turnover.
- Built a frequency-severity model of fleet loss exposure that set the retention and limit structure behind a \$60M coverage placement.

Business Analyst Chain Link Services

June 2019 – Sept 2021

- Sited warehouses by k-means clustering over demand geography, and forecast inventory demand with Prophet, cutting holding costs and stockouts.
- Restructured pick paths with an apriori model and aisle-traversal heuristics, and deployed a safety-stock rule to set offsite inventory levels.

Technical Skills

Modelling: Python, PyMC, scikit-learn, XGBoost, PyTorch, PySpark · **Optimization & simulation:** Gurobi, Pyomo, PuLP, HiGHS, SimPy, ARENA

Data & cloud: Databricks, Azure, SQL Server, DB2, Hadoop, Elasticsearch · **Shipping:** Prefect, Airflow, Git, GitHub, GitLab

Education

New Mexico State University, Las Cruces, NM — M.S. Mathematics (May 2018), B.S. Mathematics (May 2016)